Geometric Coordination and Residual RL for Robust Bimanual Handover — v2 Follow-up Study

Audit, residual diagnostics, geometry ablations and a preregistered harder v2 experiment (Isaac Lab 3.0 / Isaac Sim 6.2, PhysX)

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Summary

The v1 study found that a training-free geometry module is decisive for the two-Franka bar handover, but its frozen checkpoint selector chose the untrained policy in 8 of 9 runs, so it could not say whether residual reinforcement learning helps. This follow-up audits v1, diagnoses what the archived residual policies do, ablates the geometry module, and runs a new, harder and preregistered v2 experiment with stronger controls.

- **v1 is reproducible; two presentation/logging errors and one interpretation gap were found.** All published tables regenerate byte for byte from the downloaded raw records. The aggregate labels "1,400 in-range / 400 stress" should read 1,200 non-stress / 600 stress (E1). Recorded `episode_return` omits the terminal-step reward (E2, logging only). The v1 "untrained" policies are geometry plus a seed-specific constant offset of 1–2.5 mm, not a zero residual (E3).
- **Learned residuals on top of geometry behave like a constant bias, in the wrong direction.** In v1 and in v2 the trained combined policies output an almost state-independent offset of about −1 to −2.5 mm along the receiver's approach axis (across-case spread ≤ 0.6 mm). A tuned constant in the opposite direction (+10 mm) beats every learned policy.
- **Geometry is mostly alignment.** Alignment contributes +18.9 pp (1 cm bound) / +24.0 pp (1.5 cm) on the held-out suite, the contact-gated release +7.8 / +7.3 pp, and the shared-holding anchor 0.0 / −0.1 pp. Larger bounds help monotonically (all-case success 84.0 / 91.9 / 96.5 / 98.3% for 0.5 / 1 / 1.5 / 2 cm).
- **v2 held-out result: residual RL adds no value over strong geometry, and loses to a tuned constant.** On 2,400 fresh cases (training range ±2 cm / ±0.30 s plus ±3 cm extrapolation), combined (selected checkpoints) reaches 96.1% vs 96.5% for its own 1.5 cm geometry (−0.4 pp, 95% CI [−0.8, −0.2]) and 99.2% for geometry + 10 mm constant (−3.2 pp [−4.0, −2.3]). All differences come from the extrapolation conditions; on in-training conditions all three reach 100%.
- **Without geometry, residual RL does learn something a constant cannot.** Residual (selected) reaches 82.6% vs nominal 65.2% (+17.4 pp [+15.9, +19.0]) and nominal + 9 mm constant 69.3% (+13.3 pp [+12.0, +14.7]); its correction tracks the object offset. It remains far below geometry (96.5%).
- **The explicit contact-phase input has no effect** (combined − combined_no_phase −0.2 pp [−0.6, +0.2]).

Evidence types are kept separate below: original v1 results, the corrected v1 presentation, development diagnostics (450–1,150 development cases), v2 pilots, v2 selected checkpoints, v2 final checkpoints and untrained controls. Only the v2 final-suite numbers are held-out test evidence, and v2 was designed after seeing v1 and the development diagnostics.

1. Task, methods and questions

Two Franka Panda arms with parallel-jaw grippers hand over a rigid 20 cm bar in Isaac Lab v3.0.0 on Isaac Sim 6.2.0.0 with PhysX. A giver holds the bar; the receiver follows a fixed nominal approach-grasp schedule; the giver releases on a (perturbed) schedule. Perturbations shift the bar's start position (per-axis uniform) and the release time. Success requires two continuous seconds in which only the receiver holds the bar (giver open, no giver or table contact, small drift). Controllers act at 20 Hz through differential IK with fixed hand orientations.

method	receiver target	release
nominal	nominal schedule	scheduled
geometry	nominal + alignment (A) + shared-holding anchor (H)	contact-gated (G), 1 s cap
residual	nominal + learned residual `0.02·tanh(a)` per axis	scheduled
combined	geometry applied to nominal + learned residual	contact-gated
combined_no_phase	as combined, four explicit contact-phase inputs zeroed	contact-gated
constant controls	the learned residual replaced by a fixed offset, same composition and bound	as base

Questions: (Q1) are the v1 artifacts and conclusions reproducible, with any measurement or aggregation defects? (Q2) does a learned residual give useful state-dependent correction beyond zero, random initialization and a tuned constant? (Q3) how much do alignment, anchoring and gated release each contribute? (Q4) does the conclusion depend on the alignment bound? (Q5) under a preregistered, harder but feasible distribution, does residual learning add robust value over a strong geometry controller?

2. Audit of v1 (Q1)

Full table: docs/v2/AUDIT.md; errata: docs/v2/V1_ERRATA.md.

- **Artifacts.** The v1.0.0 assets re-downloaded anonymously match all 21 checksums; all 556 archive members re-hash correctly; checkpoint hashes, transition counts and selections are consistent with the training ledgers.
- **Numbers.** Running the v1 analysis code (clean clone at `v1.0.0`) on the downloaded raw records reproduces all 18 primary and secondary summary files byte for byte. Records: 19,800 primary and 16,200 secondary, identical case-ID sets, 0 duplicates, 0 invalid resets, 0 infrastructure errors.
- **E1 (labels).** The analysis pools conditions by the substring `stress`: 1,200 non-stress cases (clean + five in-range conditions) and 600 stress cases. Only the labels "1,400 / 400" were wrong; the corrected v1 table is in `report-v1-corrected.pdf` of the v2 release.
- **E2 (return logging).** Isaac Lab's `DirectRLEnv.step` calls `_get_dones` (where records are written) before `_get_rewards`, so `episode_return` misses the terminal step (e.g. 4.66 recorded vs 14.76 full for a clean success). PPO trained on the correct per-step rewards and no v1 result used the field; v2 records the full return.
- **Replay.** Re-running a predeclared sample of 23 cases (16 chosen at failure boundaries) for nominal, geometry and the selected + final checkpoint of all nine runs in separate processes reproduced the archived terminal reason in 456/460 records at batch size 23 and 457/460 at batch size 1. All disagreements are stress boundary cases; batched GPU physics is not bitwise reproducible across batch layouts (median per-case maximum object-position difference between batch sizes 0.04 mm, 99th percentile 9 mm).
- **Mechanics.** Action scaling (single `tanh` bound), frozen deterministic normalization at evaluation, geometry composition, release timing, timeout bootstrapping, reset retries, per-

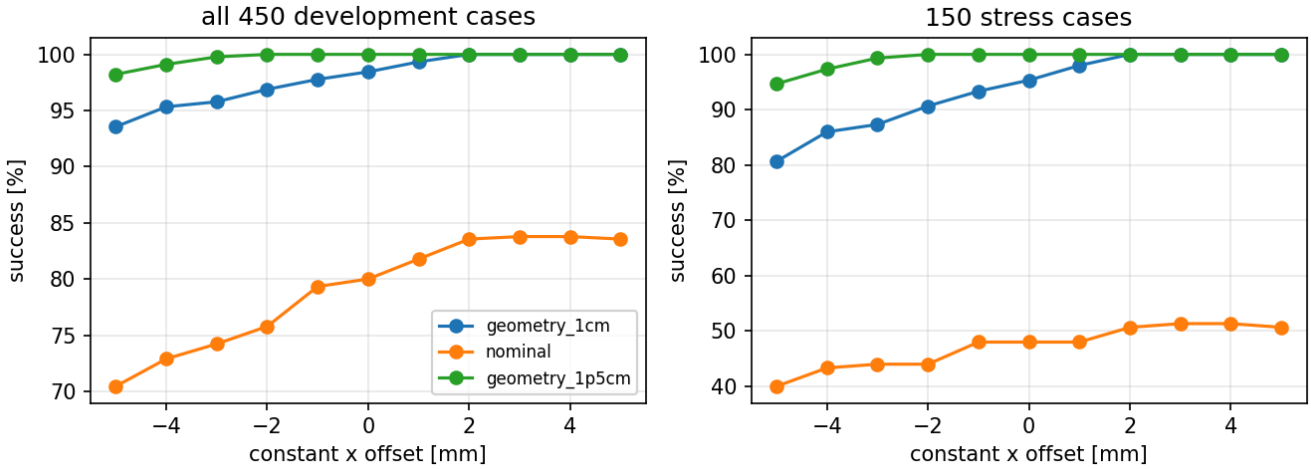
environment state isolation and the success logic were checked against the code and the installed libraries. Negative controls behave physically: with a table under the bar the receiver grips at 17 N per finger and the giver is open, but the table carries 1.3–1.6 N and the hold timer never starts (0/20 successes).

3. Residual diagnostics on development cases (Q2)

Details and traces: report/v2/diagnostics.md. On 450 fresh development cases with the v1 task configuration:

controller (development, 450 cases)	stress (150)	all (450)
geometry + zero residual	95.3	98.4
geometry + constant +3 mm (selected here)	**100.0**	**100.0**
v1 combined, untrained step 0 (3 seeds)	98.7 / 100 / 89.3	99.6 / 100 / 96.4
v1 combined, trained final (3 seeds)	88.7 / 94.7 / 93.3	96.2 / 98.2 / 97.8
nominal / nominal + 4 mm	48.0 / 51.3	80.0 / 83.8
v1 residual final (3 seeds)	52.0 / 54.0 / 61.3	84.0 / 84.7 / 87.1

All seven geometry failures have an object x offset of +17 to +19 mm (beyond the 1 cm alignment bound): the receiver inserts too deep and its palm strikes the bar end. A positive x offset shortens the insertion. The v1 combined policies learned a near-constant offset in the *negative* direction (−1.0 to −2.5 mm; across-case spread ≤ 0.4 mm; slope against the object's offset ≈ 0), and they fail precisely on cases with offsets of +15 to +19 mm. Their untrained versions are also constants (+1.4, +1.0, −2.5 mm) and their success follows the sign. The residual-only policies are state dependent (slope 0.07–0.16 against the object offset) and gain +7.8 pp on stress over nominal, but only +4.4 pp [−2.0, +11.3] over the best constant.



4. Geometry components and bounds (Q3, Q4)

Details: report/v2/geometry_ablation.md. Held-out v2 suite, residual = 0:

combination	1 cm bound: all	extrapolation	1.5 cm bound: all	extrapolation
none (= nominal)	65.2	18.5	65.3	18.7
A	83.6	49.0	89.1	57.5
G	72.5	29.3	72.6	29.3
H	65.2	18.0	65.2	18.0
A+G	92.0	74.5	96.6	86.5
A+H+G (full)	91.9	74.3	96.5	86.2

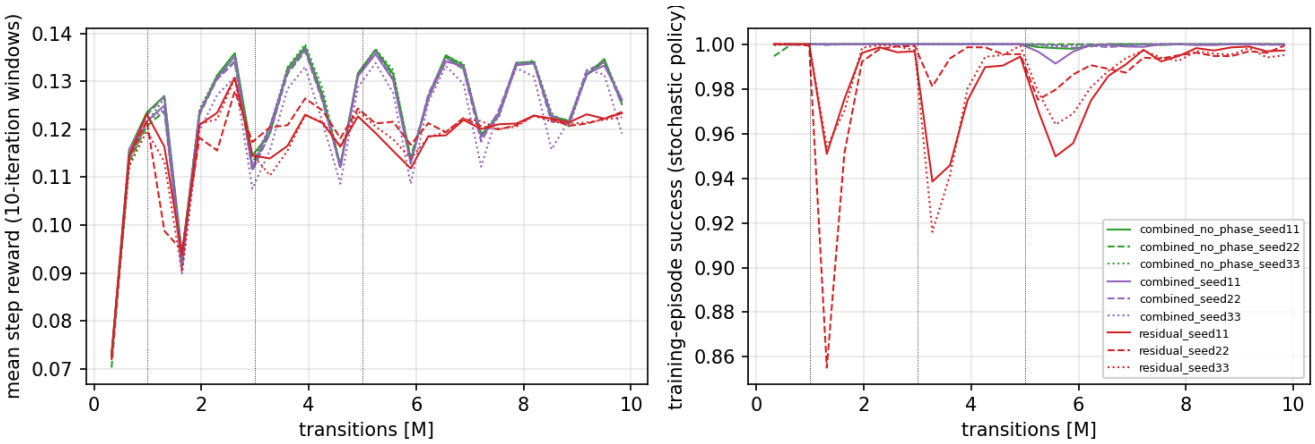
Main effects (2^3 contrasts, all cases): A +18.9 pp [+17.5, +20.3] at 1 cm and +24.0 [+22.5, +25.6] at 1.5 cm; G +7.8 [+6.9, +8.8] and +7.3 [+6.5, +8.2]; H +0.0 and −0.1. The gate's value depends on alignment: given A, it adds +8.0 pp on ±2 cm cases but +25.5 pp on ±3 cm cases (1 cm bound). The development reproduction check matched full A+H+G to v1 geometry in 449/450 cases (450/450 for "none" vs nominal). Alignment-bound sweep (all cases): 84.0 / 91.9 / 96.5 / 98.3% at 0.5 / 1 / 1.5 / 2 cm. The direction of every conclusion holds at every bound; the absolute extrapolation performance does not.

5. The v2 experiment (Q5)

Protocol: docs/v2/PROTOCOL.md (frozen at tag `experiment-v2`, protocol SHA-256 `bd79ae51…`). Main changes from v1: training curriculum reaching ±2 cm / ±0.30 s; 1.5 cm geometry in combined methods; zero-mean actor initialization; step 0 excluded from selection with new tie-breaks; 400-case stratified validation; final suite of 12 conditions × 200 fresh cases including three ±3 cm extrapolation conditions; constant controls selected on 750 development cases (nominal +9 mm, geometry 1 cm +7 mm, geometry 1.5 cm +10 mm, all along x).

Feasibility and pilots (development only). All ±3 cm resets were valid and 2 cm geometry solved 90–100%, so the extrapolation conditions were admitted. The ±2 cm / ±0.30 s training distribution is already solved by 1.5 cm geometry on development cases; this saturation was disclosed rather than changed (weakening geometry was ruled out). Two pilot configurations differing only in actor initialization were run (14.75M of the 15M pilot budget). The predeclared rule chose the zero-mean initialization (P2: 96.8% vs P1 97.6% mean success on 1,150 development cases, within the 1 pp margin). Neither combined pilot improved on its zero-residual comparator; the residual pilot improved from 62% to 90%.

Training and selection. Nine runs (residual, combined, combined_no_phase × seeds 11/22/33), each 10,027,008 transitions (8.45–8.46M of them scored, i.e. with the residual active); all five candidates per run validated on 400 cases. The rule selected checkpoints at 2M–10M; combined and combined_no_phase candidates scored 94.2–98.5% on validation (development proxy of 1.5 cm geometry: 98.5%), so several selections were decided by the declared tie-breaks.

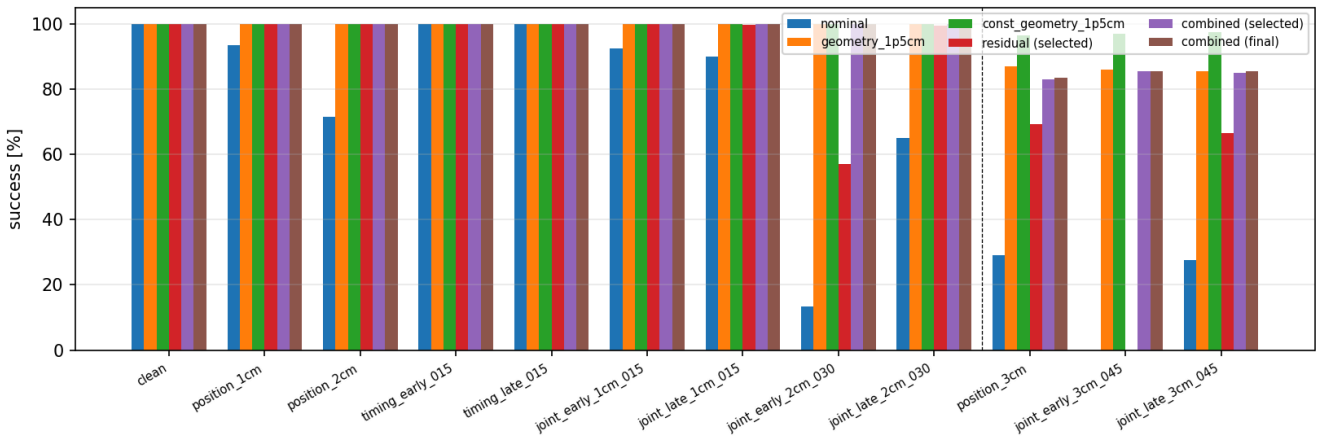


Reconciliation. 117,600 episodes were scheduled from the evaluation manifest and 117,600 recorded: 0 missing, 0 duplicates, 0 invalid resets, 0 infrastructure errors, no reset retries. Four deterministic entries are identical configurations and reuse records (e.g. ablation A+H+G at 1 cm = geometry 1 cm).

5.1 Main results (held-out v2 final suite, success % of valid resets = of scheduled cases)

controller	inner (1,200)	outer ±2 cm (600)	extrapolation ±3 cm (600)	all (2,400)
nominal	96.0	50.0	18.8	65.2
nominal + 9 mm	99.9	57.0	20.3	69.3
geometry 1 cm (v1 setting)	100.0	93.3	74.3	91.9
geometry 1.5 cm	100.0	100.0	86.2	96.5
geometry 1.5 cm + 10 mm	100.0	100.0	**97.0**	**99.2**
residual, selected (mean of 3 seeds)	99.9	85.5	45.2	82.6
residual, final	100.0	85.6	45.9	82.9
combined, selected	100.0	100.0	84.4	96.1
combined, final	100.0	100.0	84.7	96.2
combined, untrained (step 0)	100.0	100.0	86.2	96.5
combined_no_phase, selected	100.0	100.0	85.2	96.3

Per-seed all-case success: residual 82.2 / 82.8 / 82.9; combined 96.0 / 96.2 / 96.2; combined_no_phase 96.1 / 96.1 / 96.6. Per-condition results: results/v2/summary/tables.md.



5.2 Primary and declared secondary comparisons

Paired seed × case bootstrap (10,000 replicates), percentage points, all 2,400 cases; Holm-adjusted p over the five primary comparisons.

comparison	selected checkpoints	Holm p	final checkpoints
combined − geometry 1.5 cm	−0.4 [−0.8, −0.2]	0.004	−0.4 [−0.6, −0.1]
combined − (geometry 1.5 cm + 10 mm)	**−3.2 [−4.0, −2.3]**	< 0.001	−3.1 [−3.9, −2.2]
residual − nominal	**+17.4 [+15.9, +19.0]**	< 0.001	+17.7 [+16.1, +19.2]
residual − (nominal + 9 mm)	**+13.3 [+12.0, +14.7]**	< 0.001	+13.6 [+12.2, +15.0]
combined − combined_no_phase	−0.2 [−0.6, +0.2]	0.37	−0.1 [−0.3, +0.1]

By group: combined − geometry 1.5 cm is exactly 0 on inner and outer cases and −1.8 [−3.1, −0.7] on extrapolation; combined − (geometry + constant) is −12.6 [−16.0, −9.3] on extrapolation (Cohen's h −0.46). Residual − (nominal + constant) is 0.0 on inner, +28.5 [+24.6, +32.5] on outer and +24.9 [+21.7, +28.3] on extrapolation. Collision-episode rates: combined has +0.7 pp more than geometry and +4.2 pp more than geometry + constant; residual has +8.8 pp more than nominal + constant.

Untrained controls. The zero-mean untrained combined policies are behaviourally identical to 1.5 cm geometry (residual exactly 0); their records differ from the geometry records in 2 of 2,400 cases (the same two cases for all six untrained combined/no-phase policies), and the untrained residual policies differ from nominal in 10 of 2,400. This is the run-to-run nondeterminism floor of the batched simulator across evaluation processes. Trained combined policies are −0.4 pp [−0.8, −0.1] below their untrained versions.

5.3 What the learned policies do

policy (held-out)	mean signed residual x [mm]	across-case std x [mm]	slope vs object x	failures with object x > +20 mm
combined s11 / s22 / s33 (selected)	−2.4 / −2.1 / −1.2	0.3 / 0.3 / 0.6	−0.01 / +0.02 / −0.00	96/97, 92/92, 92/92
residual s11 / s22 / s33 (selected)	−1.2 / −2.9 / −1.8	2.0 / 2.1 / 1.6	+0.13 / +0.12 / +0.10	93/427, 94/414, 92/410

The v2 combined policies repeat the v1 pattern: a near-constant negative x offset, failures concentrated where the object is shifted more than 2 cm towards the receiver (beyond the 1.5 cm

alignment bound), i.e. where a positive offset is needed. PPO with the given reward does not find the beneficial direction even though it is reachable within the ±2 cm residual bound; in the training range nothing distinguishes the directions (1.5 cm geometry already succeeds), and the action-magnitude penalty favours small residuals. The residual-only policies, which face visible failures during training, learn a partial state-dependent correction (it follows ~10–13% of the object's x offset) that a constant cannot reproduce.

5.4 Failure analysis

- Nominal fails by palm collisions (deep insertion, +x offsets), timeouts from shallow grasps (−x) and drops after early release (243 drops, 496 collision episodes, 121 timeouts over 2,400 cases).
- Geometry 1.5 cm fails only beyond its bound (84 collision episodes, all in extrapolation; 1 timeout); the +10 mm constant removes all collisions (0) and leaves 18 timeouts.
- Combined fails like geometry, slightly more often (97 / 92 / 92 failures per seed, all palm collisions).
- Residual fails mainly in the joint ±2 cm / −0.30 s condition (57% success) and in extrapolation (0% on ±3 cm with 0.45 s early release: without a gate, an early release drops the bar).

Videos (10, selected by the frozen rule; every single-environment replay matched its evaluation record): matched nominal / geometry / combined comparison on `v2final-joint_early_2cm_030-0000`; alignment-only vs alignment + gate (release-gate rescue, rule fallback because no in-training drop occurred); 1 cm vs 1.5 cm geometry (alignment-bound example, palm collision vs success); a combined success and a combined failure in extrapolation; a residual success where nominal fails. Index: `videos.json` in the release.

6. Answers

- **Q1.** Yes. Every v1 table reproduces from the raw records; counts reconcile; replays agree except at failure boundaries (0.7–0.9%). Defects: count labels (E1), return logging (E2), and the "untrained" step-0 policies being constant offsets (E3). None changes a v1 number.
- **Q2.** Not on top of geometry. The trained combined residual is a constant bias in an unhelpful direction; zero, untrained and constant controls match or beat it (development and held-out). Without geometry the residual is state dependent and beats the best constant by +13.3 pp held-out.
- **Q3.** Alignment is the main component, the contact-gated release a substantial secondary one (especially for early releases and beyond the alignment bound), and the anchor has no measurable effect here.
- **Q4.** Direction no, magnitude yes: larger alignment bounds improve geometry monotonically, but residual learning never closes the gap to the stronger geometry or its constant-offset variant.
- **Q5.** No. Under the preregistered harder distribution residual learning adds no robust value over 1.5 cm geometry (−0.4 pp) and is clearly worse than geometry plus a development-tuned constant (−3.2 pp; −12.6 pp on extrapolation). Strong geometry is sufficient for this task; residual RL is useful only when geometry is absent.

7. Limitations

- Simulation only; translation perturbations with fixed orientations; one bar; privileged state and contact observations. No claim about real hardware or perception noise.
- Three training seeds per method; seed-level uncertainty is coarse (here the per-seed all-case success of the combined methods varies by at most 0.5 pp, of residual by 0.7 pp). Five primary comparisons are Holm-adjusted; the many group, condition and ablation comparisons are exploratory.
- v2 was designed after v1 and the development diagnostics (e.g. the 1.5 cm comparator, the constant controls and the extrapolation range); fresh test cases remove case reuse, not this dependence.
- The training distribution is saturated by the stronger comparator, so in-training comparisons cannot show a benefit; validation is near-saturated as well, and checkpoint selection relied on tie-breaks.
- The constant search was extended once after its grid edge was selected, and the final 1.5 cm constant is again at the edge of the extended grid (flat region, 745–746/750); a larger constant might be marginally better.
- Only one reward and PPO configuration (two pilot initializations). Other rewards (e.g. a penalty on collision proximity) might teach the beneficial offset; this was out of the assignment's bounded scope.
- Batched GPU physics is not bitwise reproducible; outcome differences between evaluation processes are 0.08–0.4% of cases and are reported.
- Videos use the Newton Warp ray tracer (A100 GPUs have no RT cores); physics is unchanged.

8. Reproduction

Commands: docs/v2/REPRODUCE.md. Source: tag `experiment-v2` (experiment code and frozen protocol); analysis, report and packaging on branch `study/v2-diagnostics` (tag in the release notes). The release contains all raw records (development, pilot, validation, final, audit replays, negative controls), every checkpoint of every v2 run and pilot, logs, videos with sidecars, and indices (`runs.json`, `checkpoints.json`, `videos.json`, `manifest.json`, `archive_inventories.json`, `SHA256SUMS`). `scripts/v2_analyze_final.py`, `scripts/v2_analyze_dev.py` and `scripts/v2_select_constants.py` regenerate every table and figure from the downloaded records.

References

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Software and AI-assistance disclosure

Software reused. NVIDIA Isaac Sim 6.2.0.0 (PhysX), Isaac Lab v3.0.0 (BSD-3-Clause; Franka Panda asset configuration, differential IK controller, contact sensors, Newton Warp renderer, RSL-RL wrapper and runner helpers), RSL-RL 5.5.1 (PPO, BSD-3-Clause), PyTorch 2.11, NumPy, Matplotlib, imageio/FFmpeg, python-pptx (presentation). Environment lock: `configs/v2/frozen/pip-freeze.txt` (identical to v1). No external datasets or demonstrations were used.

AI assistance. As in v1, the follow-up implementation, experiments, analysis code, figures, this report and the presentation were produced by an AI coding agent (Anthropic Claude, operating through Claude Code) working from the repository's written assignment on the owner's behalf, on the owner's account and compute. Every design decision is logged with UTC time and evidence in `docs/v2/DECISIONS.md`; all numbers come from the checked-in analysis scripts applied to the archived raw records. No course rubric, template or AI-use policy was supplied, so no claim of compliance with such rules is made.